ANALYSIS REPORT

CVP 1.4 h0 Offset Cross-Check

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

Cross-check of `CVP 1.4` signed model offset errors against measured `h0` / curve mean over `12416` joined candidate-curve rows and `111` candidates.

Decision

- `h0` magnitude alone does not explain most CVP 1.4 offset failures.
- Median candidate `abs(error)` versus `abs(h0)` correlation is `-0.0152`; mean is `-0.0125`.
- Median top-decile overlap lift is `1.0000` versus a random-decile baseline of `1.0`; `33` of `111` candidates reach lift `>= 2.0`.
- `94` of `111` candidates have weak absolute correlation `< 0.25`.
- Join validation is tight: maximum `CVP 1.4 truth_mean_deg - measured_h0_deg` is `0.0000` deg.

Surface Summary

Surface	Rows	Mean Abs Offset	Mean Abs h0	Abs Corr	Overlap Lift	Max Delta
`Bw`	3977	0.0044	0.0263	-0.0885	1.4824	0.0000
`Fw`	5141	0.0032	0.0585	0.0181	1.7476	0.0000
`global`	3298	0.0039	0.0424	-0.0044	1.0303	0.0000

Largest Mean Offset Error Candidates

Candidate	Rows	Mean Abs Offset	Mean Abs h0	Abs Corr	Overlap Lift
`MLP19_Fw`	97	0.0185	0.0585	-0.0388	1.0000
`harmonic_regression_global`	194	0.0180	0.0424	0.0744	1.5000
`MLP19_Bw`	97	0.0137	0.0263	-0.1186	2.0000
`rcim_retuned_XGBM19_Bw`	97	0.0101	0.0263	0.3105	5.0000
`rcim_original_MLP19_Fw`	97	0.0098	0.0585	-0.0789	0.0000
`ELM19_Bw`	97	0.0097	0.0263	-0.0313	3.0000
`rcim_retuned_MLP19_Bw`	97	0.0090	0.0263	-0.0431	1.0000
`rcim_retuned_MLP19_Fw`	97	0.0083	0.0585	-0.0620	2.0000
`rcim_retuned_ELM19_Bw`	97	0.0081	0.0263	0.3107	4.0000
`XGBM19_Bw`	97	0.0077	0.0263	-0.2231	0.0000

Strongest h0/Error Overlap Candidates

Candidate	Rows	Abs Corr	High Error	High h0	Overlap	Lift
`rcim_retuned_LGBM19_Bw`	97	0.3199	10	10	6	6.0000
`rcim_retuned_XGBM19_Bw`	97	0.3105	10	10	5	5.0000
`rcim_retuned_HGBM19_Bw`	97	0.2870	10	10	5	5.0000
`LGBM19_Fw`	97	0.0258	10	10	5	5.0000
`rcim_retuned_ELM19_Bw`	97	0.3107	10	10	4	4.0000
`XGBM19_Fw`	97	0.1168	10	10	4	4.0000
`ELM19_Fw`	97	0.3372	10	10	3	3.0000
`temporal_convolution_Bw`	97	0.0720	10	10	3	3.0000
`GBM19_Fw`	97	0.0508	10	10	3	3.0000
`residual_harmonic_lstm_sequence_sparse_rcim_Fw`	97	0.0458	10	10	3	3.0000

Strongest Absolute Correlation Candidates

Candidate	Rows	Abs Corr	Signed Corr	Mean Abs Offset	Mean Abs h0
`rcim_retuned_ET19_Fw`	97	0.3722	0.0846	0.0017	0.0585
`ELM19_Fw`	97	0.3372	-0.3204	0.0071	0.0585
`rcim_retuned_LGBM19_Bw`	97	0.3199	-0.9679	0.0073	0.0263
`rcim_retuned_ELM19_Bw`	97	0.3107	-0.5644	0.0081	0.0263
`rcim_retuned_XGBM19_Bw`	97	0.3105	-0.9868	0.0101	0.0263
`rcim_retuned_HGBM19_Bw`	97	0.2870	-0.7289	0.0034	0.0263
`rcim_original_ELM19_Fw`	97	0.2581	-0.3404	0.0050	0.0585
`rcim_retuned_RF19_Fw`	97	0.1682	0.0298	0.0012	0.0585
`rcim_original_RF19_Fw`	97	0.1618	0.0593	0.0015	0.0585
`rcim_original_XGBM19_Fw`	97	0.1571	-0.0270	0.0024	0.0585

High-Error Normal-h0 Quadrants

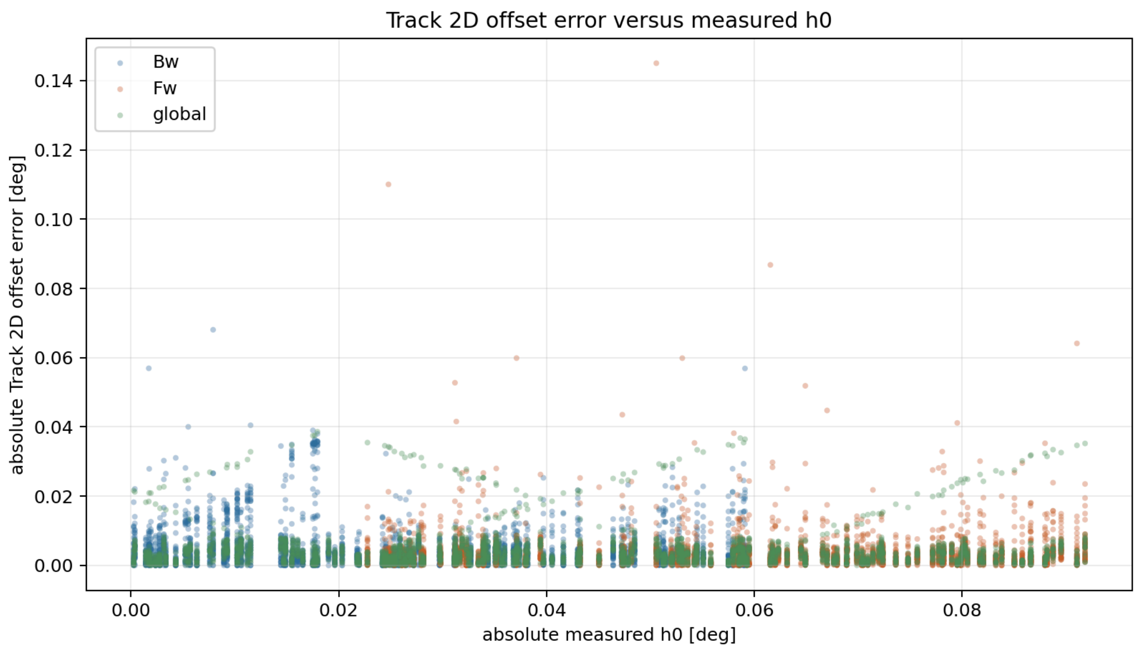
These rows are important because they contradict a pure `h0` -magnitude explanation.

Candidate	High Error + High h0	High Error + Normal h0	Normal Error + High h0
`feedforward_global`	0	20	20
`periodic_temporal_convolution_global`	1	19	19
`residual_harmonic_gru_sequence_dense240_global`	1	19	19
`residual_harmonic_lstm_sequence_dense240_global`	1	19	19
`temporal_convolution_global`	1	19	19
`gru_sequence_global`	2	18	18
`periodic_gru_sequence_global`	2	18	18
`residual_harmonic_gru_sequence_dense360_global`	2	18	18
`residual_harmonic_gru_sequence_sparse_rcim_global`	2	18	18
`residual_harmonic_lstm_sequence_dense360_global`	2	18	18

Interpretation

The cross-check supports a narrower framing: the problematic quantity is still the curve mean / `h0` channel, but the large CVP 1.4 model offset errors do not simply occur where measured `abs(h0)` is large. This points toward candidate-specific mean prediction bias, direction/regime dependence, or missing causal state information rather than a pure measured-`h0` outlier problem. The useful next diagnostic is therefore predicted-mean versus measured-`h0` surface analysis for the high-error candidates, with separate `Fw`, `Bw`, and `global` handling.

Scatter Diagnostic



CVP 1.4 offset error versus measured h0

Machine-Readable Artifacts

- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_joined_rows.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_candidate_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_surface_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_quadrant_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_summary.yaml`

Reproduction

```
powershell conda run -n pinns_env python -B scripts/reports/analysis/build_track2d_h0_offset_crosscheck_report.py
```